

Chapter 6 The Link Layer and LANs

Introduction

Terminology (術語)

Hosts and Routers : node

Layer-2 packet : frame, encapsulates datagram

Link-Layer: Context

Datagram transferred by different link protocols over different links

e.g. WiFi $\longleftrightarrow$ Ethernet Cable $\longleftrightarrow$ Cellular

Link-Layer: Services

Framing, link access :

Encapsulate datagram into frame, adding header, trailer (糾錯)

MAC addresses (in headers) :

As the IP in transport layer, in format of 48-bits binary

Provide reliable delivery between adjacent (相鄰) nodes :

Error checks in trailer

/!\ 即使 Link Layer 有糾錯, TCP 還是需要進行糾錯, 因為無法避免像 Router 重啟的狀況

Flow control

Pacing between adjacent sending and receiving nodes

Error detection :

Detect error then re-transmit or drop frame

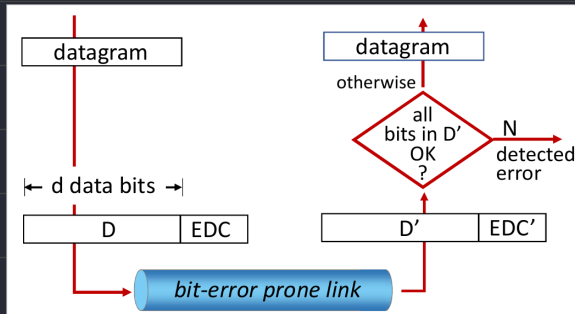
Error correction :

Correct the error without re-transmitting

Link-Layer: Implement

network interface card (NIC) (獨立網卡) or on a chip (板載網卡)

Error detection, correction



EDC: error detection and correction bits

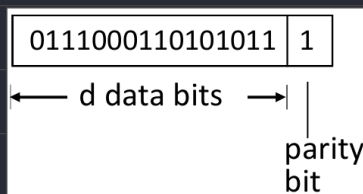
e.g. redundancy

D: data with header protected by error checkin

/!\ Not 100% reliable e.g. two bit invert

Parity checking

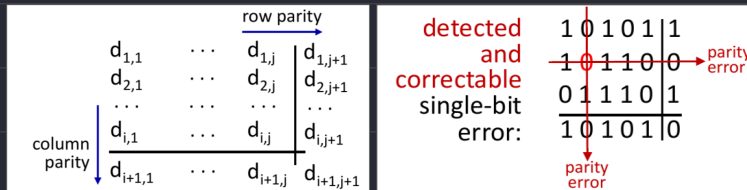
Single bit parity



Even parity: set parity to 1 if there are odd count of 1

Odd parity: set parity to 1 f there are odd count of 1

Two-dimensional bit parity

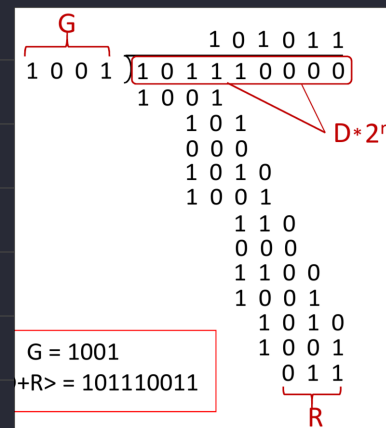


付出更多代價換更多功能

Internet checksum (review) :: SKIPPED

Cyclic Redundancy Check (CRC)

1. 設被除數D 除數G
2. 把被除數補上 $\text{number_of_bits}(G)-1$ 個 0
3. 使用長除法，但減法的部分採用 XOR，算出於數R
4. 傳送端送出內容 $f\{D\}\{R\}$
5. 接收端把 $f\{D\}\{R\}$ 除以 G 如果整除就沒錯



Multiple access protocols

An ideal multiple access protocol

一人要傳送就獨佔所有頻寬，多人就平分，去中心化，越簡單越好

MAC protocols: taxonomy (分類學)

Channel partitioning

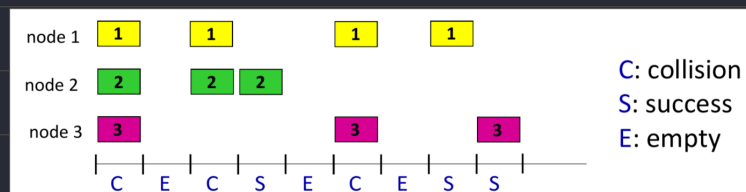
TDMA: time division multiple access

FDMA: frequency division multiple access

Random access protocols

想傳就傳，如果發生干擾（碰撞）就延遲隨機時間後重傳，期望不會再次碰撞

Slotted ALOHA

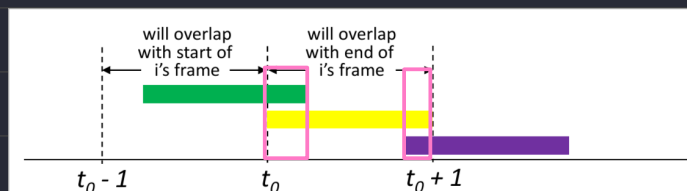


效率: $1/e = .37$

原理: 只有在時槽開始時才能傳資料，如果遇到碰撞就在下個時槽有10%的機率傳送

問題: Clock synchronization is expensive

Pure ALOHA (Unslotted Aloha)



原理: 取消時槽簡化電路，但碰撞機率更高

效率: 0.18

CSMA (carrier sense multiple access)

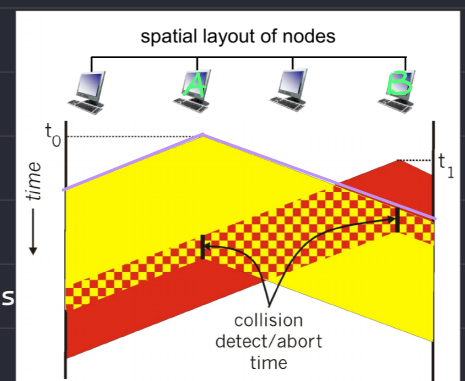
Principle: Listen before transmit

CSMA 無法完全避免碰撞，例如 B 在 A 的傳輸訊號

抵達前掃描到 channel 為空，但是等他開始送後

A 的訊號才傳播過來，此時就發生碰撞

→ Low efficiency under wireless circumstances



CSMA/CD: CSMA with collision detection Ethernet used

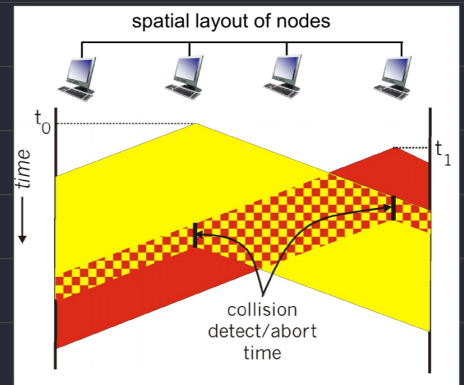
Principle

監測到碰撞後就立即停止傳輸，因為後面繼續傳也是浪費
能源與時間

Re-transmitt policy

在第M次重傳，從 $\{0, 1, 2, \dots, 2^M\}$ 中挑選一個時間 delay

道理：重傳越多次 $\rightarrow$ Channel 越繁忙 $\rightarrow$ 慢點傳



Usage

Current Ethernet use this policy to transmitt data

Problem

Not fair, if someone keep colision with others, it might wait longer

Efficiency

- T_{prop} = max prop delay between 2 nodes in LAN

- t_{trans} = time to transmit max-size frame

$$efficiency = \frac{1}{1 + 5t_{prop}/t_{trans}}$$

- efficiency goes to 1

- as t_{prop} goes to 0

- as t_{trans} goes to infinity

$\rightarrow$ Low efficiency under wireless circumstances

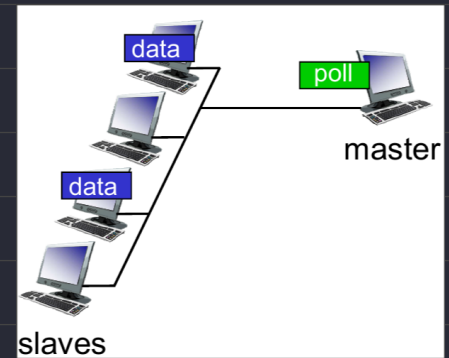
“Taking turns” MAC protocols

Polling

Master: Ask everyone if it want to transmitt

Slave: Only transmitt data if master asks

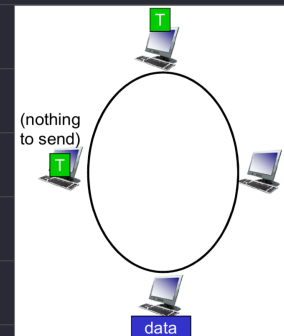
Concerns : Polling overhead, latency, master dead
(額外開銷)



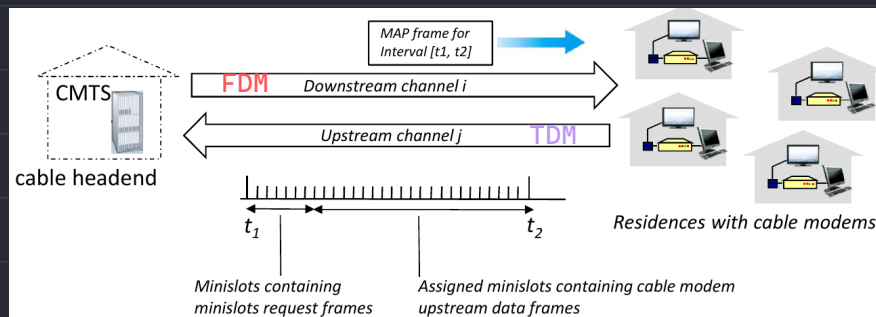
Token passing

Control token: passed from one node to next client.

Concerns: token overhead, latency, token miss



Cable access network



DOCSIS: data over cable service interface specification

LANs

Addressing, ARP

IP address (32-bit): Used in the network layer (WAN)

MAC addresses (48-bit): Used in the local network (LAN)

A permanent address allocation administered by IEEE

MAC flat address (portability): The address not changed when moved to other LAN

ARP: address resolution protocol

ARP table

Definition

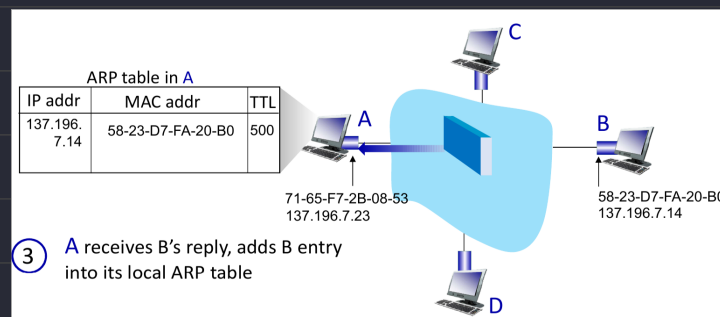
Data Structure: A list of < IP address; MAC address; TTL >

TTL (Time To Live): when address mapping will be forgotten (20min)

Protocol in action

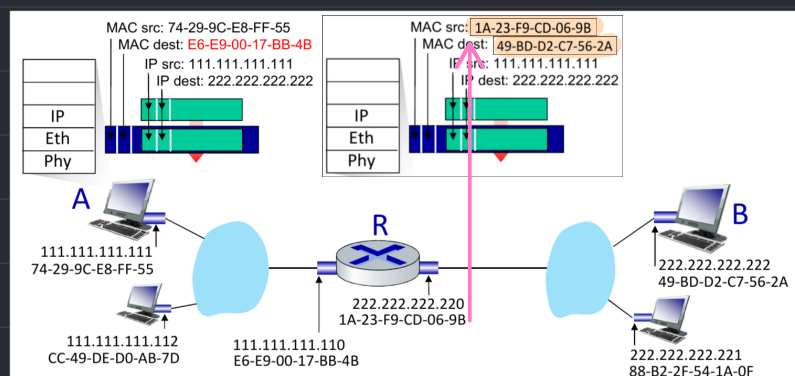
Case 1: Under the same router

1. A broadcasts ARP query (dest MAC address = FF-FF-FF-FF-FF-FF)
2. All nodes on LAN receive ARP query and respond IP MAC pair to sender



Case 2: Under different subnet

1. Use DHCP and ARP to get router's MAC address
2. Router will forward the packet to the destination back



Ethernet

Physical topology

Bus: popular through mid 90s

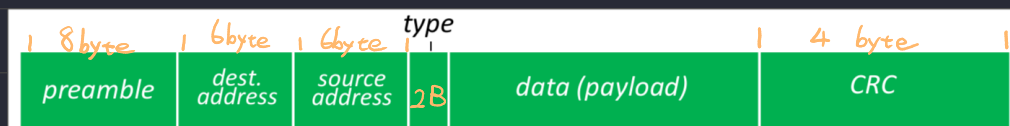
All nodes in same collision domain (CSMA/CD)

Switched: prevails today

Switch can resolve all collide problems, but ethernet still use CDMA/CD



Ethernet frame structure



preamble: 7 byte of 10101010 followed by one byte of 10101011

type: indicates higher layer protocol

mostly IP but others possible, e.g., Novell IPX, AppleTalk

used to demultiplex up at receiver

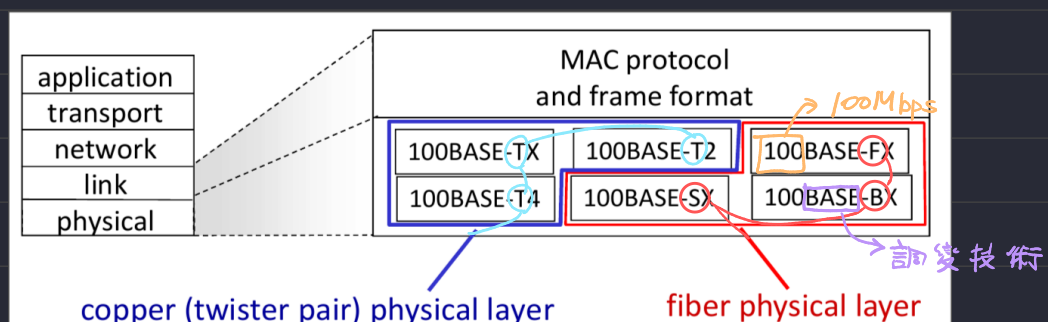
CRC: cyclic redundancy check at receiver

error detected: frame is dropped

Ethernet: unreliable, connectionless

1. Connectionless: No logic connection before data transfer
2. Unreliable: Packet dropped when CRC failed, make upper to handle error
3. Ethernet's MAC protocol: unslotted CSMA/CD with binary backoff

802.3 Ethernet standards: link & physical layers



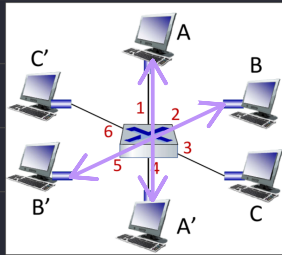
Switches

A link-layer device: forward packets by it's MAC address

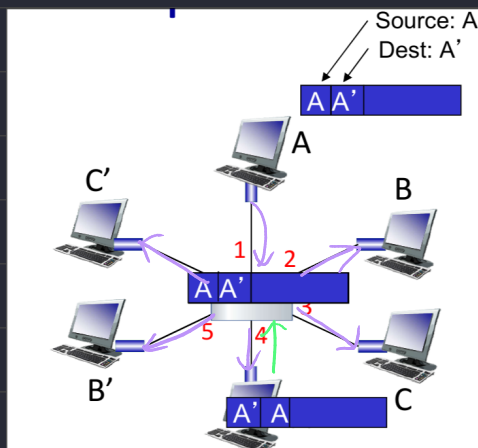
Transparent: hosts unaware of presence of switches

Plug-and-play, self-learning: do not need to be configured

A-to-A' and B-to-B' can transmit simultaneously (同時), without collisions



Self-learning



1. 收到封包 → 紀錄來源的 MAC 與 Link 對應關係
2. 如果轉發表沒有目標 MAC → 直接 broadcast
3. 等待目標回應後，紀錄 MAC 與 Link 對應關係

MAC addr	interface	TTL	switch table (initially empty)
A	1	60	
A'	4	60	

Switches vs. routers

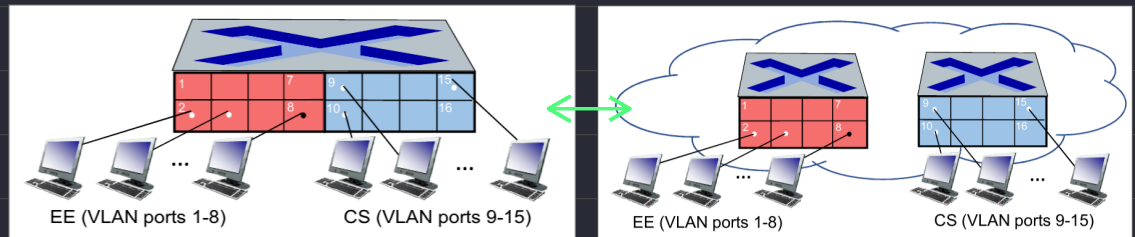
	Switches	Router
store-and-forward	Yes, in link layer (MAC)	Yes, in network layer (IP)
forwarding tables	Yes, from self learning	Yes, from routing algorithms

VLANs (Virtual LANs)

Motivation

1. 使用 Switch 會讓封包在整個網路中廣播（第一次），造成資源浪費與安全問題
2. 使用者一定要使用接入的 switch 來決定進入的網路

Port-based VLANs



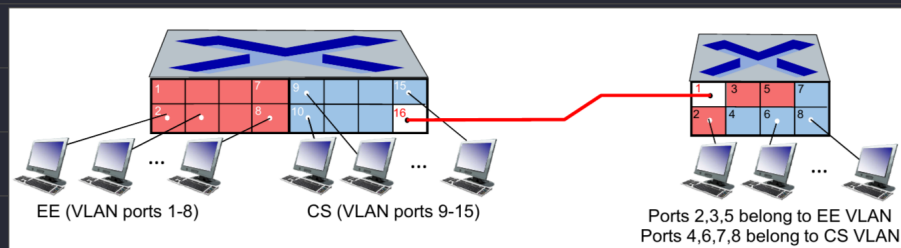
traffic isolation: Port 1~8 will be isolated from port 9~15

dynamic membership: VLAN port can be configured to different port

forwarding between VLANs: done via routing (just as with separate switches)

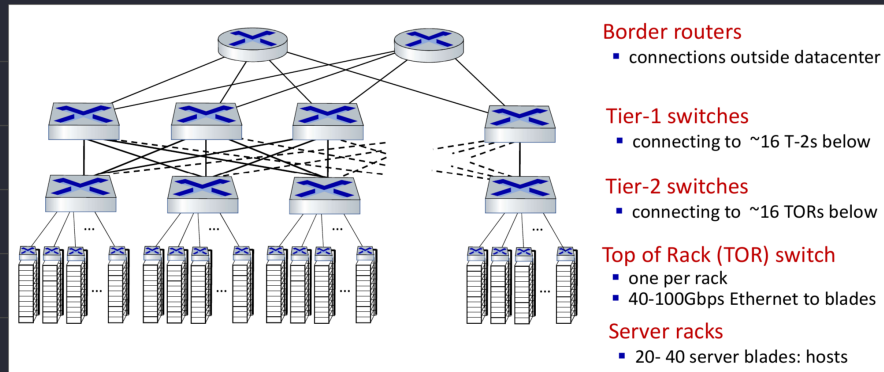
trunk port:

carries frames between VLANs defined over multiple physical switches



Data center networking

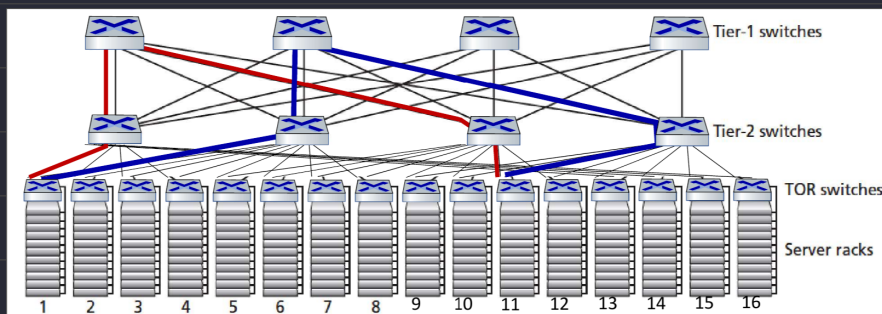
Datacenter networks



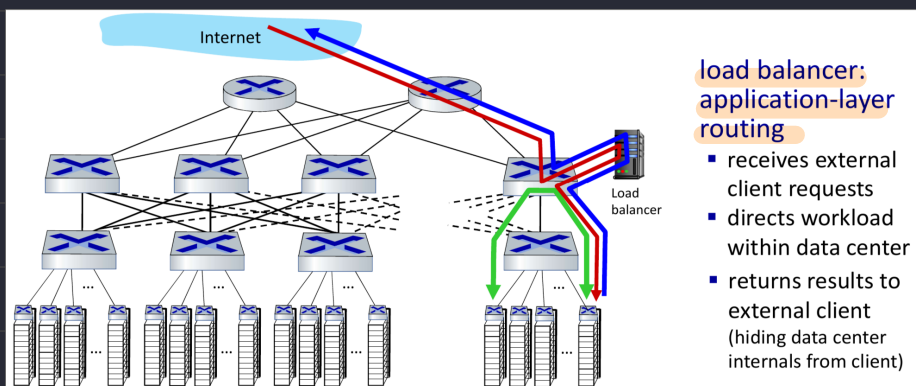
Multipath

Rich interconnection among switches (adding redundant connection)

- increased throughput between racks (multiple routing paths possible)
- increased reliability via redundancy



Application-layer routing



Protocol innovations

SKIP.